Unit: Unit 0: Foundations of Algebra
Topic: GCF LCM

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Find the GCF of 12 and 18 using prime factorization.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q2. What is the LCM of 9 and 12 using multiples?
(A) 12
(B) 18
(C) 27
(D) 36
(E) 54

Q3. Find the GCF of 20 and 30 using factors.
(A) 1
(B) 2
(C) 5
(D) 10
(E) 20

Q4. A bookshelf has 12 shelves, and each shelf can hold 8 books. What is the LCM of 12 and 8?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48

Q5. Find the LCM of 15 and 25 using prime factorization.
(A) 15
(B) 25
(C) 50
(D) 75
(E) 125

Q6. A group of friends want to share some candy equally. If they have 18 pieces of candy and there are 3 friends, what is the GCF of 18 and 3?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9

Q7. Find the GCF of 24 and 30 using prime factorization.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q8. What is the LCM of 4 and 9 using multiples?
(A) 4
(B) 9
(C) 12
(D) 18
(E) 36

Open Ended Questions (FRQ)

Q9. Tom has 18 boxes of pens, and each box contains 12 pens. If Alex has 24 boxes of pencils, and each box contains 15 pencils, what is the LCM of the total number of pens Tom has and the total number of pencils Alex has? Show your work.

Q10. A bakery is making a special batch of cookies for a holiday sale. They need to package 48 cookies into boxes that hold 8 cookies each, and 60 cookies into boxes that hold 12 cookies each. What is the GCF of the total number of boxes of cookies they will have? Explain your reasoning.

Chef's Tips

- Make sure students show their work for FRQs.
- Emphasize the importance of using prime factorization or multiples to find GCF and LCM.
- Encourage students to check their answers by using different methods.